



Clamp-on Flow Meter FD-R Series IO-Link Instruction Manual



This instruction manual explains the IO-Link specifications for the clamp-on flow meter FD-R Series (hereinafter abbreviated as FD-R).

Details about the FD-R functions and how to use the unit can be found in the "FD-R Series Instruction Manual". A copy of the "FD-R Series Instruction Manual" can be downloaded off of the KEYENCE website. Alternatively, contact a local KEYENCE representative.

The settings file can also be downloaded off of the KEYENCE website.

If it is not possible to gain access to the internet to download the file, contact a local KEYENCE representative.

KEYENCE Website: <http://www.keyence.com>

1. Specifications

Model	FD-R50 FD-R80 FD-R125 FD-R200
IO-Link approved standard	v. 1.1
Baud rate	COM2 (38.4kbps)
Min. cycle time	3.2 ms
Process data length	4 Byte

2. Wiring

M12 connector



Pin No.	Symbol	Color	Name
1	L+	Brown	20 - 30 VDC
2	Other/-	White	Selectable*1
3	L-	Blue	0V
4*2	Q1/QA/C	Black	Output*3/ IO-Link*4

*1 One of the following values is assigned: not used, output 2, external input, or analog output.

*2 Assignment changes according to the settings on the IO-Link master.

*3 When IO-Link is not connected, output 1 or analog output is assigned.

*4 When connecting with IO-Link, do so with control output assigned.

Failing to do so may lead to damage to the master or to the FD-R Series unit.

3. Process data

Process data is a function that communicates specific data in a constant cycle.

The format in which the process data is output, can be selected from the 5 types below.

Refer to "Index 146" in "Device Parameter" on page 3, or "5-10 Process Data Output Format" in "5. Settings" on page 6 for information on how to select the format type.

● Process Data Structure: 0 = Instantaneous flow rate + Temperature(1C) + Stability level + Output2 + Output1 (Default)

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Instantaneous flow rate*1, 2															
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Temperature (1 unit = 1°C/1°F)*3										0	0	Stability level*4		Output 2	Output 1

● Process Data Structure: 1 = Instantaneous flow rate + Temperature(0.1C) + Output2 + Output1

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Instantaneous flow rate*1, 2															
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Temperature (1 unit = 0.1°C/0.1°F)*5													0	Output 2	Output 1

● Process Data Structure: 2 = Instantaneous flow rate

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous flow rate*1, 2															

● Process Data Structure: 3 = Total accumulated flow(one-tenth) + Output2 + Output1

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Total accumulated flow (1 unit = 10 L/10 gal/10 m³)*6															
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Total accumulated flow (1 unit = 10 L/10 gal/10 m³)*6													0	Output 2	Output 1

● Process Data Structure: 4 = Total accumulated flow

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Total accumulated flow (1 unit = 1 L/1 gal/1 m³)*7															
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Total accumulated flow (1 unit = 1 L/1 gal/1 m³)*7															

*1 If the flow meter displays something other than a value, one of the following will be output.

Flow meter display	Output
ERC	Normal
ERE	Normal
ERP	Normal
REV	0
SETCLOCK	Normal
---	0

*2 The range of the instantaneous flow rate is 0 to 50000. The unit is linked to indexes 205/233 of the service data.

*3 The range of the temperature is -40 to 140. The unit is linked to index 209 of the service data. Negative temperature values are subtracted from 512 and the resulting values are output.

*4 The stability of the ultrasonic detection is indicated through one of five level: 0 to 4. The higher the stability level, the more stable detection becomes.

*5 The range of the temperature is -401 to 1401. The unit is linked to index 209 of the service data. Negative temperature values are subtracted from 8192 and the resulting values are output.

*6 The range of total accumulated flow is 0 to 99999999. The unit is linked to index 205 of the service data.

*7 The range of total accumulated flow is 0 to 999999999. The unit is linked to index 205 of the service data.

4. Service Data

Service data is a function that refreshes information between the IO-Link master and slave only when there is a send request. Reading/writing of the flow meter settings and status, and execution of operation commands (tuning, etc.) can be carried out.

● Meaning of each item

Item	Value	Meaning
Item	-	Shows the notation in the instruction manual
Format	Record	Data structure
	Bit (x)	Bit No. x in Record
	String	Text string
	UInt	Unsigned integer
	Int	Signed integer
Access	R	Read possible
	R/W	Read/write possible
	W	Write possible
	C	Operation command
Data storage	√	Compatible with the data storage function*1
	-	Not compatible with the data storage function

*1 Flow meter settings are backed up with the IO-Link master. This allows settings to be restored if the flow meter is replaced.

NOTICE	If parameter writing is executed from IO-Link during initialization, the flow meter may behave in an unexpected manner. Do not write parameters while initialization is being executed.
---------------	--

Point When a value that is greater than or equal to the FD-R's set upper limit is entered, the FD-R operates with the set upper limit. Also, when a value that is less than or equal to the FD-R's set lower limit is entered, the FD-R operates with the set lower limit.

4-1 Parameter List

Predefined parameter

Index	Item	Name	Format	Length	Access	Data storage	Default value	Range	Remark
12 (0x0C)	Device Access Locks	Device Access Locks	Record	2 Byte	R/W	-	-	-	-
2 (0x02)	Data Storage Lock*1	Data Storage Lock	Bit (1)	1 Bit	-	-	0	0 = Unlock 1 = Lock	-
16 (0x10)	Vendor Name	Vendor Name	String	64 Byte	R	-	Keyence		-
17 (0x11)	Vendor Text	Vendor Text	String	64 Byte	R	-	www.keyence.com		-
18 (0x12)	Product Name	Product Name	String	64 Byte	R	-	(Differs for each model)*2		-
19 (0x13)	Product ID	Product ID	String	64 Byte	R	-	(Differs for each model)*2		-
20 (0x14)	Product Text	Product Text	String	64 Byte	R	-	Clamp-on Flow Meter		-
22 (0x16)	Hardware Version	Hardware Version	String	64 Byte	R	-	-		-
23 (0x17)	Firmware Version	Firmware Version	String	64 Byte	R	-	-		-
24 (0x18)	Application Specific Tag	Application Specific Tag	String	32 Byte	R/W	√	***	-	-
37 (0x25)	Detailed Device Status	Detailed Device Status	Record	12 Byte	R	√	0	0xFF91 = Data Storage Upload Request 0x1800 = Reverse flow - The fluid flows in the opposite direction of the setting 0x1801 = Clock error - Set current date and time settings again 0x5000 = Device hardware fault - Device Exchange 0x7710 = Short circuit - Check installation	Refer to "6. Events"
40 (0x28)	Process Data Input	Process Data Input	UInt	4 Byte	R	-	-	-	-

*1 Backing up and restoring in the data storage function can be locked.

*2 The following models are applicable: FD-R50/FD-R80/FD-R125/FD-R200

Device Parameter

Index	Name	Reference item	Format	Length	Access	Data storage	Default value	Range	Remark
100 (0x64)	Selecting the ch.2 functions	Selecting the ch.2 function (refer to 5-C)	UInt	1 Byte	R/W	√	0	0 = Not used 1 = Control output 2 = External input 3 = Analog output	-
101 (0x65)	Selecting NPN/PNP	Selecting NPN/PNP (refer to 5-D)	UInt	1 Byte	R/W	√	0	0 = NPN 1 = PNP	-
102 (0x66)	Response time	Response time (refer to 5-10)	UInt	1 Byte	R/W	√	3	0 = 0.5 sec 1 = 1.0 sec 2 = 2.5 sec 3 = 5.0 sec 4 = 10.0 sec 5 = 30.0 sec 6 = 60.0 sec 7 = 120.0 sec 8 = 200.0 sec	-
103 (0x67)	Analog output current	Analog output current (refer to 5-6)	UInt	1 Byte	R/W	√	0	0 = 4 to 20 mA output 1 = 0 to 20 mA output	Linked to Indexes 105 to 109
104 (0x68)	Command request	Command request	UInt	1 Byte	C	-	-	0 = Reset integrated flow 1 = Perform conditional monitoring 2 = Cancel conditional monitoring 3 = Reset flow hold values 4 = Reset temperature hold values 5 = Perform origin adjustment 6 = Cancel origin adjustment	-
105 (0x69)	Analog lower flow limit	Setting the flow analog output range ³ (refer to 5-8)	UInt	4 Byte	R/W	√	0	0 to 99999	Linked to Indexes 103/109 Unit : Linked to Indexes 205/233
106 (0x6A)	Analog upper flow limit		UInt	4 Byte	R/W	√	FD-R50:6000 FD-R80:1500 FD-R125:3700 FD-R200:9500	0 to 99999	
107 (0x6B)	Analog lower temperature limit	Setting the temperature analog output range ⁴ (refer to 5-9)	Int	2 Byte	R/W	√	-200	-32768 to 32767	Linked to Indexes 103/109
108 (0x6C)	Analog upper temperature limit		Int	2 Byte	R/W	√	1200	-32768 to 32767	Unit : 0.1, linked to Index 209
109 (0x6D)	Analog selection	Selecting the analog output ⁵ (refer to 5-7)	UInt	1 Byte	R/W	√	0	0 = Flow analog output 1 = Temperature analog output	Linked to Indexes 103 and 105 through 108
110 (0x6E)	Output 1 logic	Output 1 logic (refer to 5-2)	UInt	1 Byte	R/W	√	0	0 = N.O. 1 = N.C.	-
111 (0x6F)	Output 1 detection mode	Output 1 detection mode (refer to 5-1)	UInt	1 Byte	R/W	√	0	0 = Instantaneous flow rate mode 1 = Area mode 2 = Pulse output mode 3 = Integrated flow mode	-
112 (0x70)	Output 1 flow set value/Lower flow limit value 1	Output 1 flow set value/Output 1 lower flow limit value	UInt	4 Byte	R/W	√	FD-R50:600 FD-R80:150 FD-R125:370 FD-R200:950	0 to 99999	Unit : linked to Indexes 205/233
113 (0x71)	Upper flow limit value1	Output 1 upper flow limit value ⁶	UInt	4 Byte	R/W	√	FD-R50:3600 FD-R80:900 FD-R125:2200 FD-R200:5700	0 to 99999	
118 (0x76)	Output 2 logic	Output 2 logic ⁷ (refer to 5-4)	UInt	1 Byte	R/W	√	0	0 = N.O. 1 = N.C.	-
119 (0x77)	Output 2 detection mode	Output 2 detection mode ⁷ (refer to 5-3)	UInt	1 Byte	R/W	√	0	0 = Instantaneous flow rate mode 1 = Area mode 2 = Temperature alarm mode 3 = Error output mode 4 = Error + temperature alarm mode	-
120 (0x78)	Output 2 flow set value/Lower flow limit value 2	Output 2 flow set value/Output 2 lower flow limit value ⁸	UInt	4 Byte	R/W	√	FD-R50:1200 FD-R80:300 FD-R125:740 FD-R200:1900	0 to 99999	Unit : linked to Indexes 205/233
121 (0x79)	Upper flow limit value 2	Output 2 upper flow limit value ⁹	UInt	4 Byte	R/W	√	FD-R50:3000 FD-R80:750 FD-R125:1800 FD-R200:4700	0 to 99999	
122 (0x7A)	Lower temperature limit value	Temperature lower limit ¹⁰	Int	2 Byte	R/W	√	0	-32768 to 32767	Units: 0.1, linked to Index 209
123 (0x7B)	Upper temperature limit value	Temperature upper limit ¹⁰	Int	2 Byte	R/W	√	1000	-32768 to 32767	
129 (0x81)	Selecting the input function	Selecting the input function ¹¹ (refer to 5-5)	UInt	1 Byte	R/W	√	0	0 = Integrated flow reset input 1 = Flow rate zero input 2 = Origin adjustment input	-
131 (0x83)	Hysteresis	Hysteresis (refer to 5-15)	UInt	4 Byte	R/W	√	FD-R50:60 FD-R80:15 FD-R125:37 FD-R200:95	0 to 99999	Linked to Indexes 112/113/120/121 Unit : linked to Indexes 205/233
138 (0x8A)	Power-saving mode	Power-saving mode (refer to 5-21)	UInt	1 Byte	R/W	√	0	0 = Normal status 1 = Power-saving status	-
139 (0x8B)	Key lock	Key lock	UInt	1 Byte	R/W	√	0	0 = Unlock 1 = Lock	Linked to Index 140
140 (0x8C)	Key lock method	Key lock method (refer to 5-25)	UInt	1 Byte	R/W	√	0	0 = Normal (without password) 1 = Password-protected key lock	Linked to Indexes 139/141

141 (0x8D)	Key lock password	Password for password-protected key lock	UInt	2 Byte	R/W	√	0	0 to 9999	Linked to Indexes 139/140
142 (0x8E)	Additional functions menu	Additional functions menu (refer to 5-12)	UInt	1 Byte	R/W	√	0	0 = Settings complete 1 = Additional functions menu	-
144 (0x90)	User tag1	User tag* ¹²	UInt	4 Byte	R/W	√	0	0 to 4294967295	-
145 (0x91)	User tag2		UInt	2 Byte	R/W	√	0	0 to 65535	-
146 (0x92)	Process data structure	Process data output format	UInt	1 Byte	R/W	√	0	0 = Instantaneous flow rate + Temperature(1C) + Stability level + Output2 + Output1 1 = Instantaneous flow rate + Temperature(0.1C) + Output2 + Output1 2 = Instantaneous flow rate 3 = Total accumulated flow (one-tenth) + Output2 + Output1 4 = Total accumulated flow	-
148 (0x94)	Instantaneous flow	Instantaneous flow rate	UInt	4 Byte	R	-	-	0 to 99999	Unit : linked to Indexes 205/233
149 (0x95)	Total accumulated flow	Total accumulated flow* ¹³	UInt	4 Byte	R	-	-	0 to 4294967295	-
150 (0x96)	Temperature	Temperature	Int	2 Byte	R	-	-	-32768 to 32767	Units: 0.1, linked to Index 209
160 (0xA0)	Peak hold flow	Flow rate peak value	UInt	4 Byte	R	-	-	0 to 99999	Unit : linked to Indexes 205/233
161 (0xA1)	Bottom hold flow	Flow rate bottom value	UInt	4 Byte	R	-	-	0 to 99999	
162 (0xA2)	Peak hold temperature	Temperature peak value	Int	2 Byte	R	-	-	-32768 to 32767	Units: 0.1, linked to Index 209
163 (0xA3)	Bottom hold temperature	Temperature bottom value	Int	2 Byte	R	-	-	-32768 to 32767	
180 (0xB4)	Stability level	Stability level* ¹⁴	UInt	1 Byte	R	-	-	0 = Detection not possible 1 = Low 2 = Medium-low 3 = Medium-high 4 = High	-
181 (0xB5)	Origin adjustment	Identify if origin adjustment is currently being used* ¹⁵	UInt	1 Byte	R	-	0	0 = OFF 1 = ON	-
182 (0xB6)	Conditional monitoring function	Conditional monitoring display	UInt	1 Byte	R	-	0	0 = OFF 1 = ON	-
189 (0xBD)	Simulation temperature value	Simulated temperature (refer to 5-24)	Int	2 Byte	R/W	-	0	-32768 to 32767	Linked to Index 220 Unit: 0.1, linked to Index 209
190 (0xBE)	Simulation flow value	Simulated flow rate (refer to 5-23)	UInt	4 Byte	R/W	-	0	0 to 99999	Linked to Index 220 Unit: linked to Indexes 205/233
191 (0xBF)	Reference flow rate	Reference point of conditional monitoring function	UInt	4 Byte	R	-	0	0 to 99999	Unit: linked to Indexes 205/233
192 (0xC0)	Zero cut flow rate	Zero cut flow rate (refer to 5-16)	UInt	4 Byte	R/W	√	FD-R50:360 FD-R80:90 FD-R125:220 FD-R200:570	0 to 99999	Unit: linked to Indexes 205/233
193 (0xC1)	Adjusting flow rate span	Flow rate span adjustment* ¹⁶ (refer to 5-28)	UInt	2 Byte	R/W	√	1000	100 to 10000	Unit: 0.001
194 (0xC2)	Integrated flow set value	Set value for integrated flow* ¹⁷	UInt	4 Byte	R/W	√	FD-R50:600 FD-R80:1500 FD-R125:3700 FD-R200:9500	0 to 99999	Linked to Index 203
195 (0xC3)	Detection hold time	Detection hold time (refer to 5-17)	UInt	1 Byte	R/W	√	1	0 = 0.5sec 1 = 1.0sec 2 = 2.5sec 3 = 5.0sec 4 = 10.0sec 5 = 30.0sec 6 = 60.0sec	-
196 (0xC4)	Integrated flow pulse weight	Integrated pulse weight* ¹⁸ (refer to 5-18)	UInt	4 Byte	R/W	√	100	2 to 99999	0.01L, 0.001m ³ , 0.01gal linked to Index 205
200 (0xC8)	Correcting the flow rate value	Correcting the flow rate value (refer to 5-27)	UInt	1 Byte	R/W	√	0	0 = No correction 1 = Correcting various values 2 = Setting flow rate magnification (flow rate span adjustment)	-
201 (0xC9)	Selecting the pipe thickness	Selecting the pipe thickness* ¹⁹ (refer to 5-31)	UInt	1 Byte	R/W	√	0	0 = SGP 1 = Sch20S 2 = Sch40 3 = Sch80 4 = Pipe thickness input	-
202 (0xCA)	Standard pipe size	Selecting the standard pipe size (refer to 5-H)	UInt	1 Byte	R/W	√	0	When FD-R50 is used: 0 = 1 1/2", 1 = 2" When FD-R80 is used: 0 = 2 1/2", 1 = 3" When FD-R125 is used: 0 = 4", 1 = 5" When FD-R200 is used: 0 = 6", 1 = 8"	-

203 (0xCB)	Integrated flow unit	Integrated flow unit* ¹⁷ (refer to 5-11)	UInt	1 Byte	R/W	√	1	0 = 0.1m ³ / 0.1gal 1 = 1L / 1m ³ / 1gal 2 = 10L / 10m ³ / 10gal 3 = 100L / 100m ³ / 100gal 4 = 1000L / 1000m ³ / 1000gal 5 = 10000L / 10000gal	Linked to Indexes 194/205
204 (0xCC)	Flow direction	Selecting the flow direction (refer to 5-E)	UInt	1 Byte	R/W	√	0	0 = From left to right 1 = From right to left	-
205 (0xCD)	Flow units	Selecting the flow unit (refer to 5-F)	UInt	1 Byte	R/W	√	0	0 = L/min 1 = m ³ /h 2 = gal/min	-
206 (0xCE)	Correcting the outer diameter	Correcting the outer diameter* ¹⁹ (refer to 5-29)	UInt	1 Byte	R/W	√	0	0 = No correction 1 = Correction	-
207 (0xCF)	Outer diameter of pipe input value	Pipe outer diameter input* ¹⁹ (refer to 5-30)	UInt	2 Byte	R/W	√	FD-R50:605 FD-R80:891 FD-R125:1398 FD-R200:2163	1 to 65535	Linked to Index 206 Unit: 0.1 [mm]
208 (0xD0)	Pipe thickness input value	Pipe thickness input* ¹⁹ (refer to 5-32)	UInt	2 Byte	R/W	√	FD-R50:38 FD-R80:42 FD-R125:45 FD-R200:58	1 to 300	Linked to Index 201 Unit: 0.1 [mm]
209 (0xD1)	Temperature units	Selecting the temperature unit (refer to 5-G)	UInt	1 Byte	R/W	√	0	0 = Celsius (°C) 1 = Fahrenheit (°F)	-
210 (0xD2)	Liquid type	Selecting the liquid type* ¹⁹ (refer to 5-33)	UInt	1 Byte	R/W	√	0	0 = Water 1 = Value input	-
213 (0xD5)	Ultrasonic speed	Ultrasonic speed input* ¹⁹ (refer to 5-34)	UInt	2 Byte	R/W	√	1497	Other than the FD-R200: 1000 to 1700 FD-R200: 1200 to 1700	Linked to Index 210 Unit: 1 [m/s]
214 (0xD6)	Viscosity	Kinematic viscosity input* ¹⁹ (refer to 5-35)	UInt	4 Byte	R/W	√	10	1 to 99999	Linked to Index 210 Unit: 0.1 [mm ² /s,cSt]
215 (0xD7)	Measurement unit expansion	Expanding the measurement units (refer to 5-I)	UInt	1 Byte	R/W	√	0	0 = OFF 1 = ON	Linked to Indexes 205/209
220(0xDC)	Simulation mode	Simulation mode (refer to 5-22)	UInt	1 Byte	R/W	-	0	0 = OFF 1 = ON	Linked to Indexes 189/190
221 (0xDD)	Flow rate zero input	Flow rate zero input	UInt	1 Byte	R/W	-	0	0 = OFF 1 = ON	-
230 (0xE6)	Display indicator illumination mode	Display indicator illumination mode (refer to 5-19)	UInt	1 Byte	R/W	√	0	0 = Green light - light off 1 = light off - Red light 2 = Green light - Red light 3 = PMI (predictive maintenance information) 4 =OFF	-
231 (0xE7)	Display averaging	Display averaging (refer to 5-14)	UInt	1 Byte	R/W	√	1	0 to 10	-
232 (0xE8)	Display resolution	Display resolution (refer to 5-13)	UInt	1 Byte	R/W	√	2	0 = 0.01 1 = 0.1 2 = 1	-
233 (0xE9)	Flow resolution	Unit of setting* ²⁰	UInt	1 Byte	R	-	2	0 = 0.01 1 = 0.1 2 = 1	-
234 (0xEA)	Display brightness	Display brightness (refer to 5-20)	UInt	1 Byte	R/W	√	0	0 = Standard (indoor use) 1 = High (outdoor use)	-
253 (0xFD)	Command status	Command status	UInt	1 Byte	R	-	0	0 = Command input idle 1 = Command input exe 2 = Command input success 3 = Command input failure	-

- *3 Valid only when "Analog output" is selected for index 100 "Selecting ch.2 function" and then "Flow analog output" is selected for index 109 "Selecting the analog output".
- *4 Valid only when "Analog output" is selected for index 100 "Selecting ch.2 function" and then "Temperature analog output" is selected for index 109 "Selecting the analog output".
- *5 Valid only when "Analog output" is selected for index 100 "Selecting ch.2 function".
- *6 Valid only when "Area mode" is selected for index 111 "Output 1 detection mode".
- *7 Valid only when "Control output" is selected for index 100 "Selecting ch.2 function".
- *8 Valid only when "Control output" is selected for index 100 "Selecting ch.2 function" and then "Instantaneous flow rate mode" or "Area mode" is selected for index 119 "Output 2 detection mode".
- *9 Valid only when "Control output" is selected for index 100 "Selecting ch.2 function" and then "Area mode" is selected for index 119 "Output 2 detection mode".
- *10 Valid only when "Control output" is selected for index 100 "Selecting ch.2 function" and then "Temperature alarm mode" or "Error + temperature alarm mode" is selected for index 119 "Output 2 detection mode".
- *11 Valid only when "External input" is selected for index 100 "Selecting ch.2 function".
- *12 Flow meter specific number can be allocated.
- *13 The unit is linked to index 205. (When "L" or "gal" is selected: 1 [L, gal]. When "m³" is selected: 0.1 [m³].)
- *14 Indicates level of stability of the ultrasonic detection. The higher the stability level, the more stable detection becomes.
- *15 Indicates change in zero position from factory setting.
- *16 Valid only when "Setting flow rate magnification (flow rate span adjustment)" is selected for index 200 "Correcting flow rate value".
- *17 Valid only when "Integrated flow mode" is selected for index 111 "Output 1 detection mode".
- *18 Valid only when "Pulse output mode" is selected for index 111 "Output 1 detection mode".
- *19 Valid only when "Correcting various values" is selected for index 200 "Correcting flow rate value".
- *20 This displays the resolution of the data related to the flow rate.

Standard Command

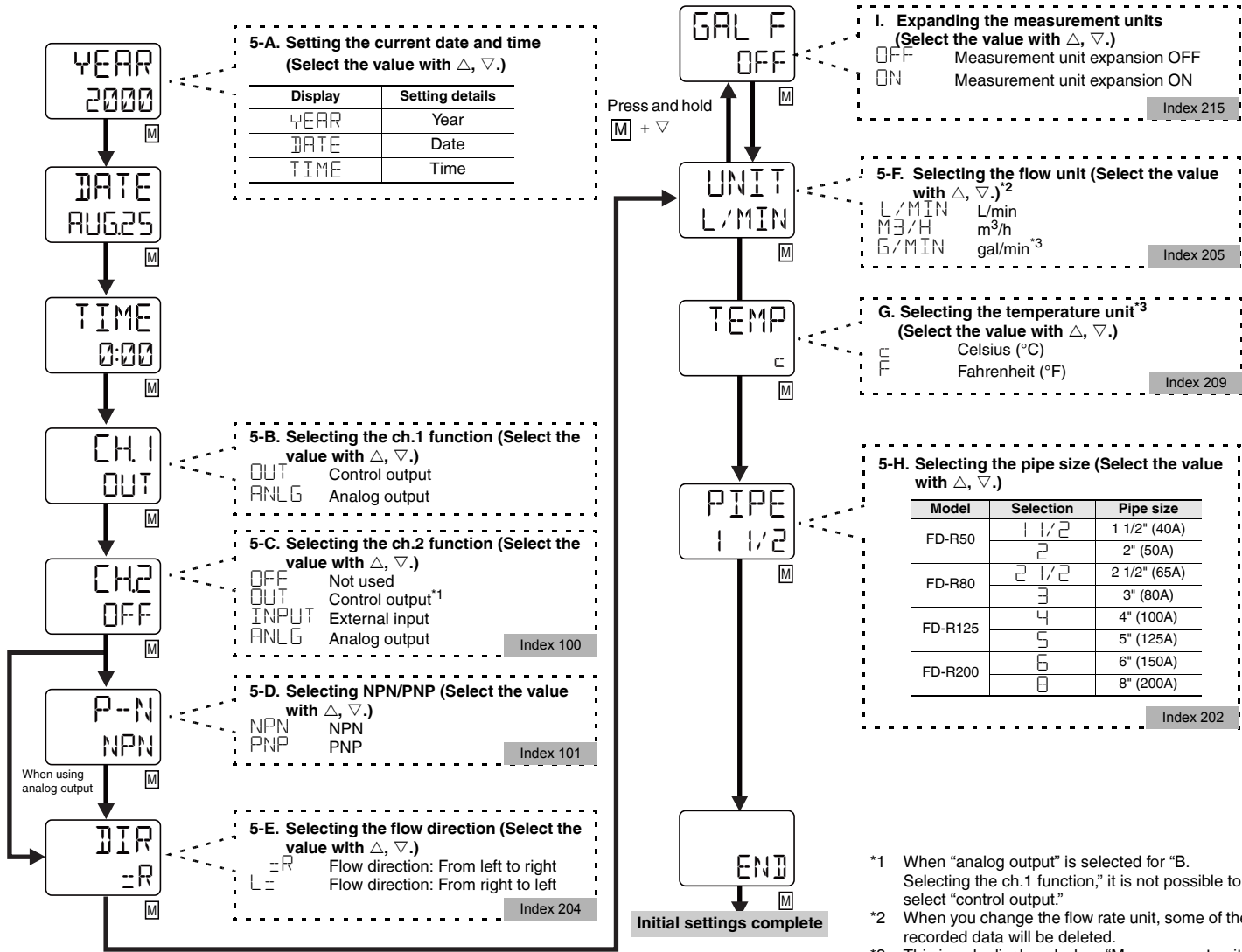
Index	Name	Reference item	Format	Length	Access	Data storage	Default value	Range	Remark
2 (0x02)	System Command	Initialization	UInt	1 Byte	W	—	130	Initialization	-

5. Settings

Some of the items that can be set in "4-1 Parameter List" on page 2 can also be set using the flow meter itself.
This section shows how to configure certain settings using the flow meter.
Refer to "Setting Descriptions" in the FD-R Series Instruction Manual for details.

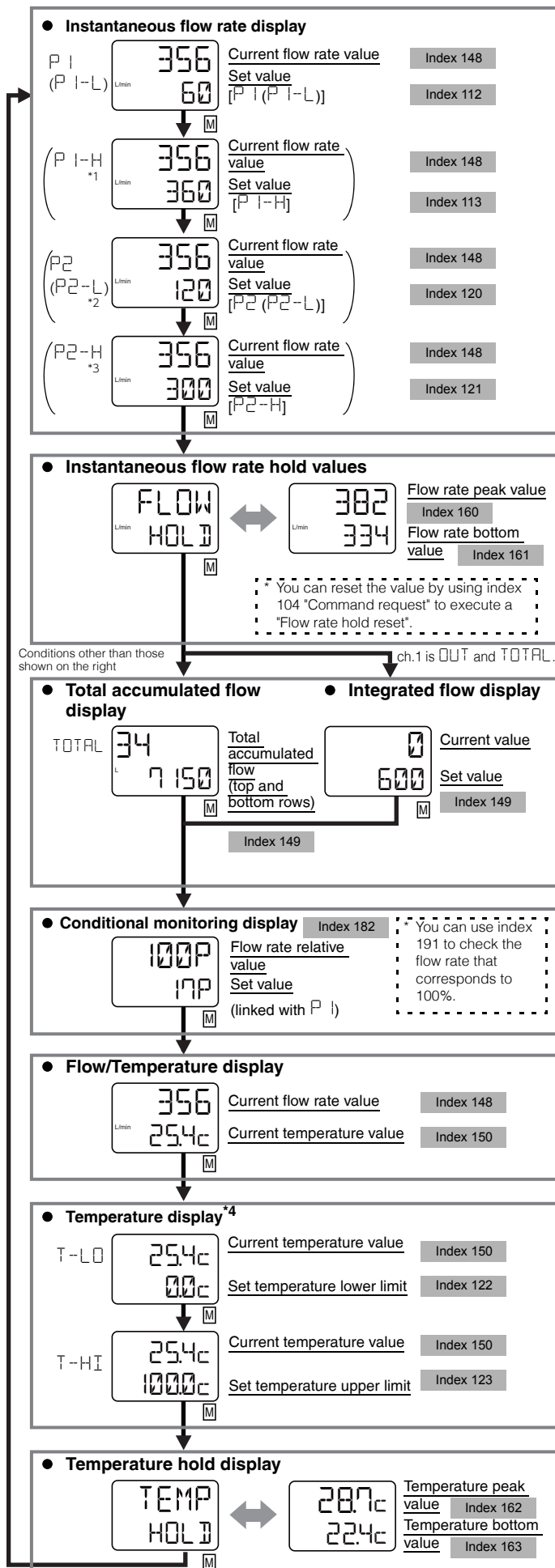
Initial Settings

When IO-Link is connected, set ch.1 to OUT.



*1 When "analog output" is selected for "B. Selecting the ch.1 function," it is not possible to select "control output."
*2 When you change the flow rate unit, some of the recorded data will be deleted.
*3 This is only displayed when "Measurement unit expansion ON" is selected for "I. Expanding the measurement units".

Switching the Display and Configuring Settings

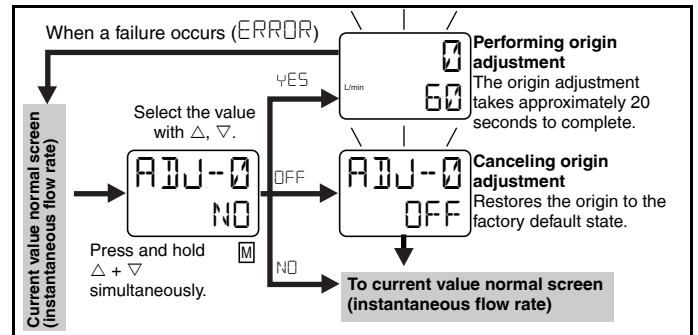


- *1 This is only valid when "Area mode" is selected for index 111 "Output 1 detection mode".
- *2 This is only valid when "Control output" is selected for index 100 "Selecting the ch.2 functions" and "Instantaneous flow rate mode" or "Area mode" is selected for index 119 "Output 2 detection mode".
- *3 This is only valid when "Control output" is selected for index 100 "Selecting the ch.2 functions" and "Area mode" is selected for index 119 "Output 2 detection mode".
- *4 This is only valid when "Control output" is selected for index 100 "Selecting the ch.2 functions" and "Temperature alarm mode" or "Error output + temperature alarm mode" is selected for index 119 "Output 2 detection mode".

Useful Functions

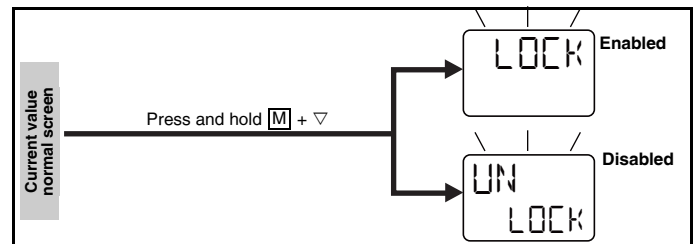
Origin Adjustment Index 104

Point Perform this function when the pipe is filled with fluid and the fluid is not moving. Performing this function when the pipe is not filled with fluid or in other such situations in which detection is unstable may lead to failures (ERROR).

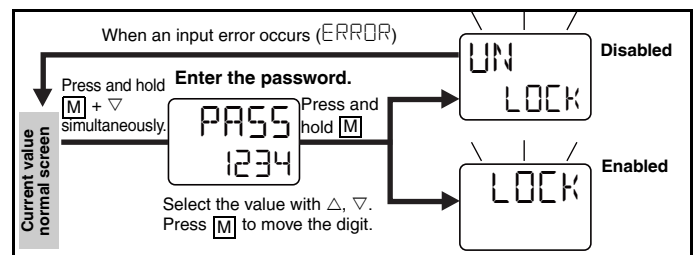


Key Lock

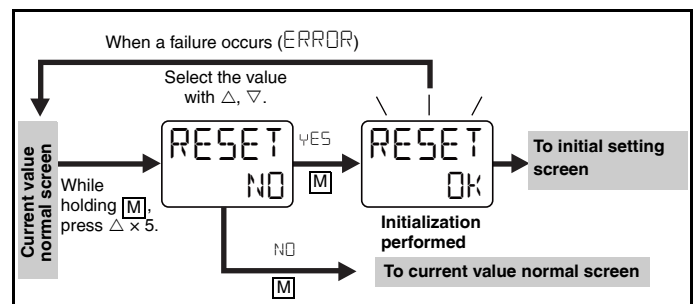
Enabling/disabling the key lock Index 139



Enabling/disabling the password-protected key lock Index 140/141



Initialization Index 2

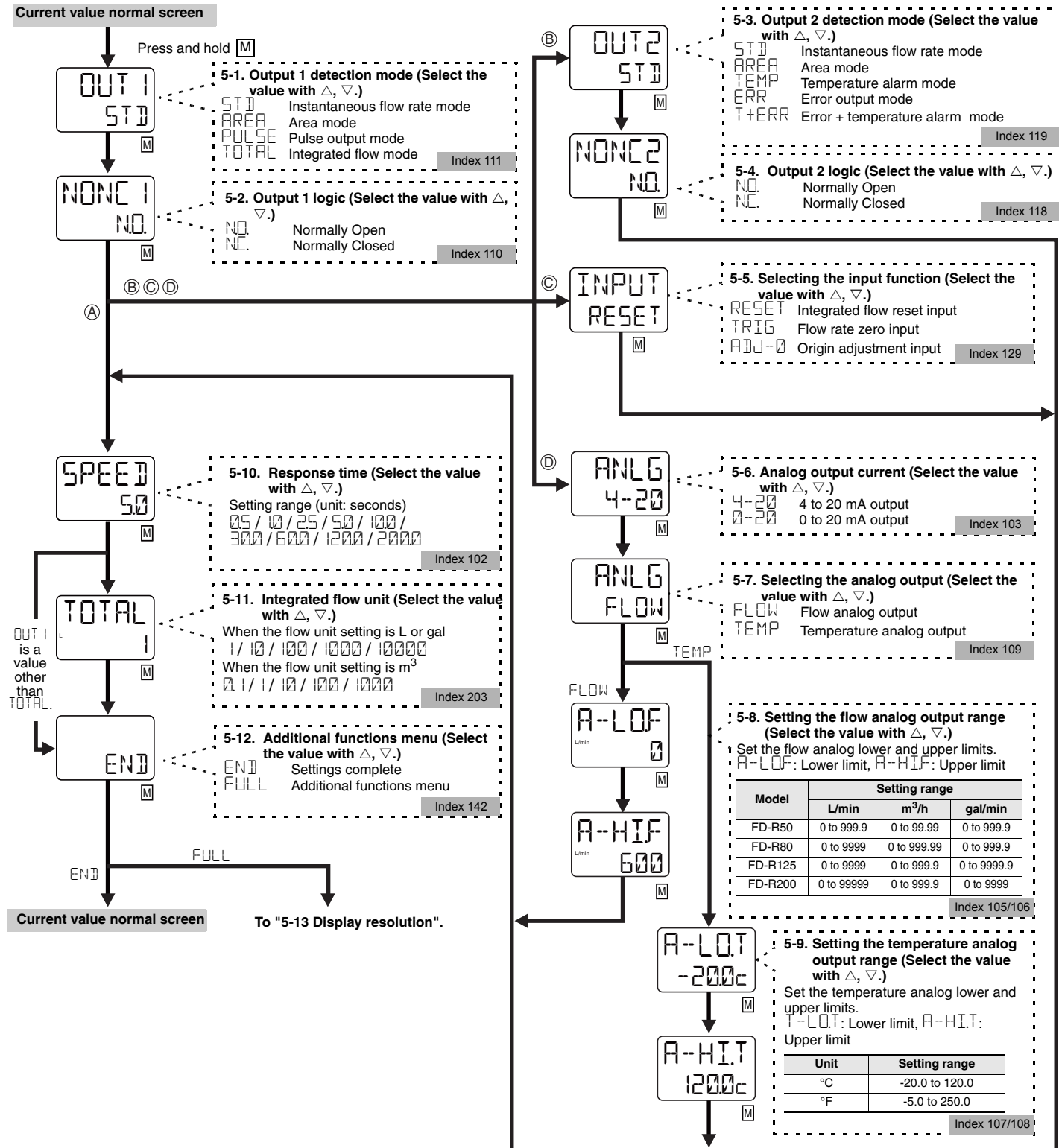


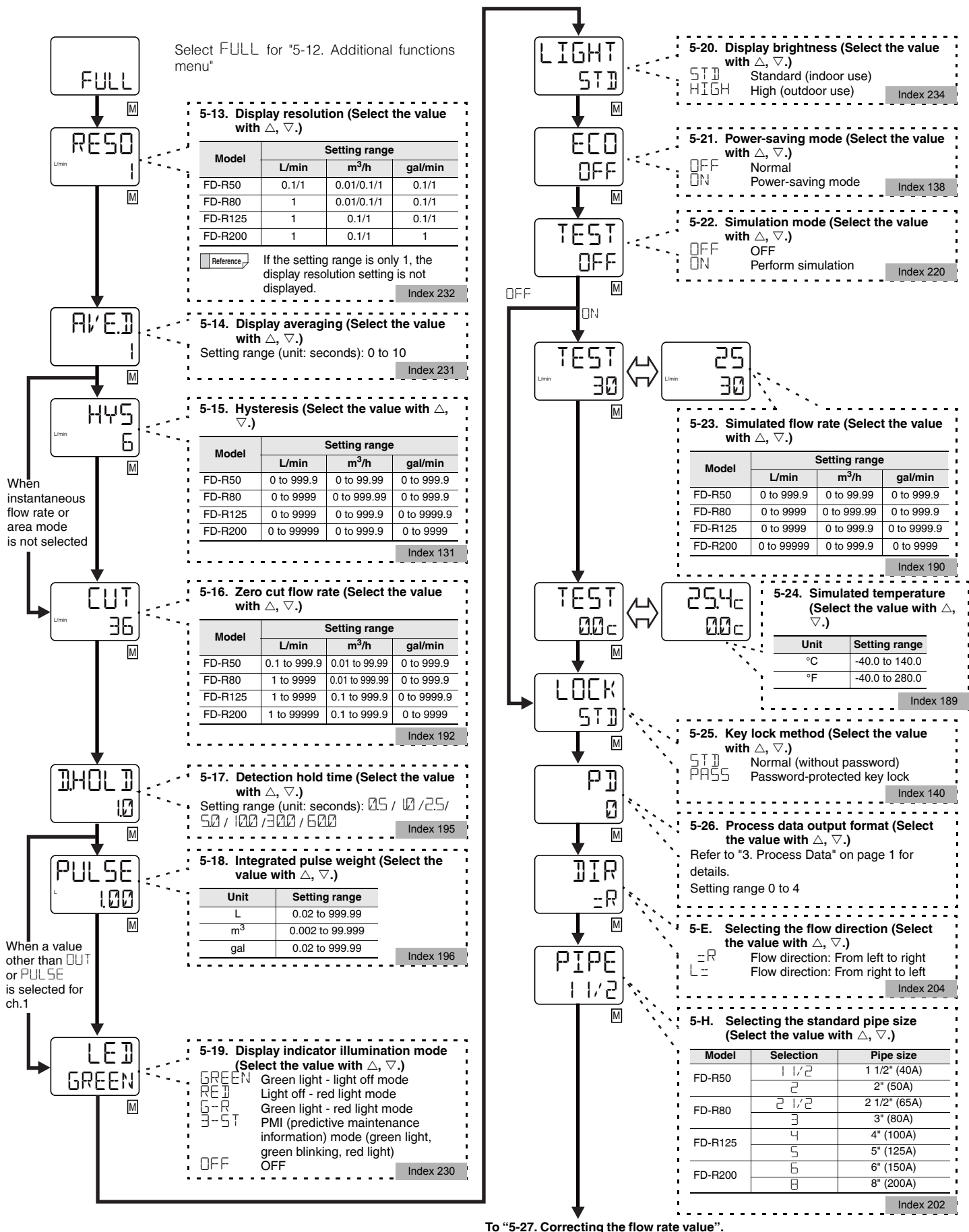
Detailed Settings

When IO-Link is connected, set ch.1 to OUT.

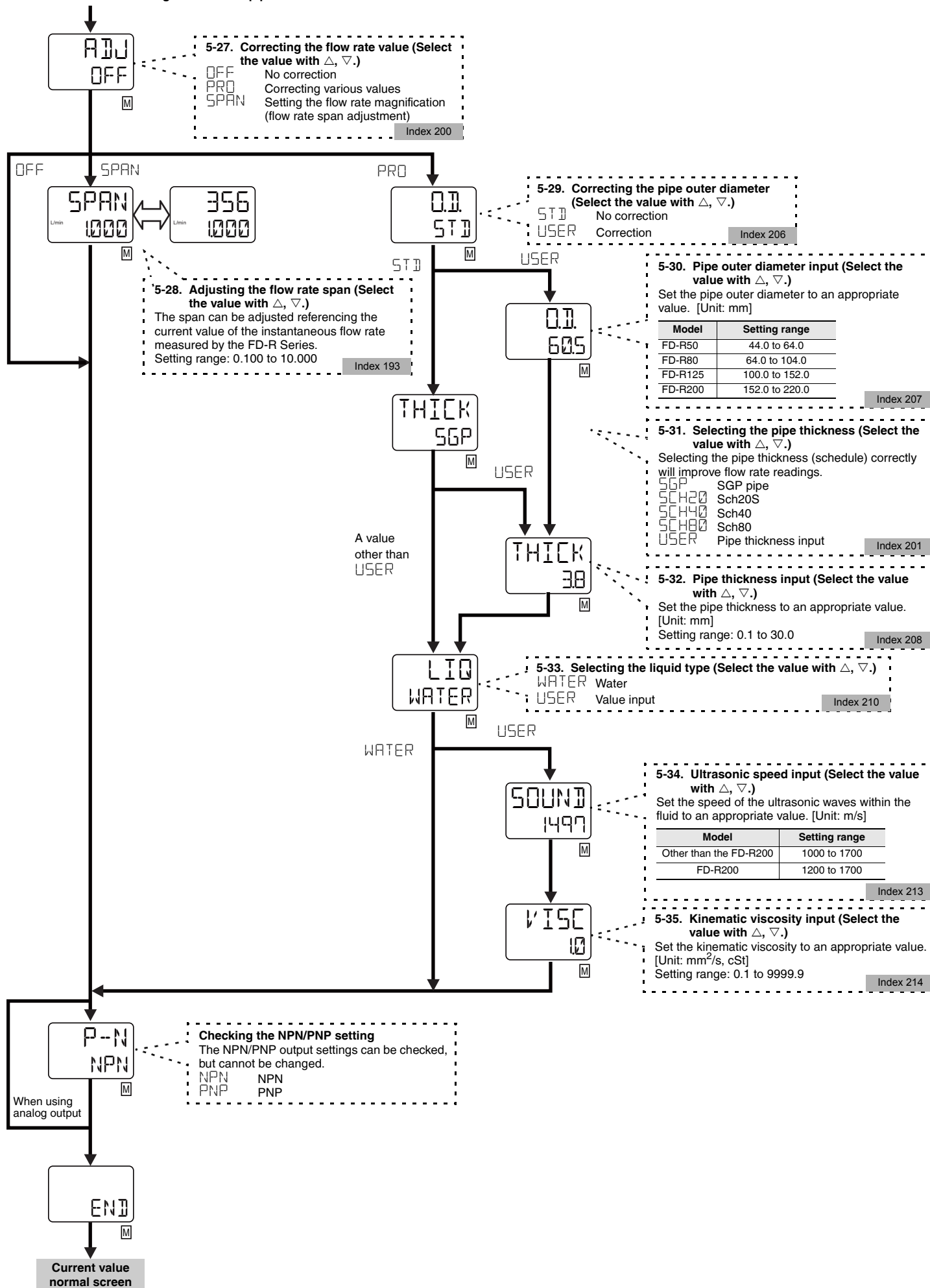
The setting items vary depending on index 100 "Selecting the ch.2 functions".

Pattern	Ch.1 setting	Ch.2 setting
(A)	OUT	OFF
(B)	OUT	OUT
(C)	OUT	INPUT
(D)	OUT	ANLG





From "5-G. Selecting the standard pipe size".



6. Events

When an event occurs, the flow meter turns ON the event flag in the process data telegram.

The master detects the event flag and reads the value that corresponds to the event that has occurred.

This allows the status of the flow meter to be forwarded to the PLC or monitor through the IO-Link master.

Service data exchange is not carried out while an event is being read.

Default value	Name	Meaning
0xFF91	Data Storage Upload Request	Settings were changed on the flow meter.
0x1800	Reverse flow - The fluid flows in the opposite direction of the setting	The fluid flows in the opposite direction of the setting.
0x1801	Clock error - Set current date and time settings again	The clock battery module has been fully discharged. Reset the time.
0x5000	Device hardware fault - Device Exchange	An error (ErE) has occurred on the flow meter.
0x7710	Short circuit - Check installation	An overcurrent error (ErC) has occurred.

WARRANTIES AND DISCLAIMERS

- (1) KEYENCE warrants the Products to be free of defects in materials and workmanship for a period of one (1) year from the date of shipment. If any models or samples were shown to Buyer, such models or samples were used merely to illustrate the general type and quality of the Products and not to represent that the Products would necessarily conform to said models or samples. Any Products found to be defective must be shipped to KEYENCE with all shipping costs paid by Buyer or offered to KEYENCE for inspection and examination. Upon examination by KEYENCE, KEYENCE, at its sole option, will refund the purchase price of, or repair or replace at no charge any Products found to be defective. This warranty does not apply to any defects resulting from any action of Buyer, including but not limited to improper installation, improper interfacing, improper repair, unauthorized modification, misapplication and mishandling, such as exposure to excessive current, heat, coldness, moisture, vibration or outdoors air. Components which wear are not warranted.
- (2) KEYENCE is pleased to offer suggestions on the use of its various Products. They are only suggestions, and it is Buyer's responsibility to ascertain the fitness of the Products for Buyer's intended use. KEYENCE will not be responsible for any damages that may result from the use of the Products.
- (3) The Products and any samples ("Products/Samples") supplied to Buyer are not to be used internally in humans, for human transportation, as safety devices or fail-safe systems, unless their written specifications state otherwise. Should any Products/Samples be used in such a manner or misused in any way, KEYENCE assumes no responsibility, and additionally Buyer will indemnify KEYENCE and hold KEYENCE harmless from any liability or damage whatsoever arising out of any misuse of the Products/Samples.
- (4) **OTHER THAN AS STATED HEREIN, THE PRODUCTS/SAMPLES ARE PROVIDED WITH NO OTHER WARRANTIES WHATSOEVER. ALL EXPRESS, IMPLIED, AND STATUTORY WARRANTIES, INCLUDING, WITHOUT LIMITATION, THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND NON-INFRINGEMENT OF PROPRIETARY RIGHTS, ARE EXPRESSLY DISCLAIMED.**
IN NO EVENT SHALL KEYENCE AND ITS AFFILIATED ENTITIES BE LIABLE TO ANY PERSON OR ENTITY FOR ANY DIRECT, INDIRECT, INCIDENTAL, PUNITIVE, SPECIAL OR CONSEQUENTIAL DAMAGES (INCLUDING, WITHOUT LIMITATION, ANY DAMAGES RESULTING FROM LOSS OF USE, BUSINESS INTERRUPTION, LOSS OF INFORMATION, LOSS OR INACCURACY OF DATA, LOSS OF PROFITS, LOSS OF SAVINGS, THE COST OF PROCUREMENT OF SUBSTITUTED GOODS, SERVICES OR TECHNOLOGIES, OR FOR ANY MATTER ARISING OUT OF OR IN CONNECTION WITH THE USE OR INABILITY TO USE THE PRODUCTS, EVEN IF KEYENCE OR ONE OF ITS AFFILIATED ENTITIES WAS ADVISED OF A POSSIBLE THIRD PARTY'S CLAIM FOR DAMAGES OR ANY OTHER CLAIM AGAINST BUYER. In some jurisdictions, some of the foregoing warranty disclaimers or damage limitations may not apply.

BUYER'S TRANSFER OBLIGATIONS:

If the Products/Samples purchased by Buyer are to be resold or delivered to a third party, Buyer must provide such third party with a copy of this document, all specifications, manuals, catalogs, leaflets and written information provided to Buyer pertaining to the Products/Samples.

E 1101-3

KEYENCE CORPORATION

1-3-14, Higashi-Nakajima, Higashi-Yodogawa-ku,
Osaka, 533-8555, Japan
PHONE: +81-6-6379-2211

www.keyence.com

AUSTRIA Ph: +43 22 36-3782 66-0	HONG KONG Ph: +852-3104-1010	NETHERLANDS Ph: +31 40 20 66 100	TAIWAN Ph: +886-2-2721-8080
BELGIUM Ph: +32 1 528 1222	HUNGARY Ph: +36 1 802 73 60	PHILIPPINES Ph: +63-(0)2-981-5000	THAILAND Ph: +66-2-369-2777
BRAZIL Ph: +55-11-3045-4011	INDIA Ph: +91-44-4963-0900	POLAND Ph: +48 71 36861 60	UK & IRELAND Ph: +44-1908-696900
CANADA Ph: +1-905-366-7655	INDONESIA Ph: +62-21-2966-0120	ROMANIA Ph: +40 269-232-808	USA Ph: +1-201-930-0100
CHINA Ph: +86-21-3357-1001	ITALY Ph: +39-02-6688220	SINGAPORE Ph: +65-6392-1011	VIETNAM Ph: +84-4-3772-5555
CZECH REPUBLIC Ph: +420 222 191 483	KOREA Ph: +82-31-789-4300	SLOVAKIA Ph: +421 2 5939 6461	
FRANCE Ph: +33 1 56 37 78 00	MALAYSIA Ph: +60-3-7883-2211	SLOVENIA Ph: +386 1-4701-666	
GERMANY Ph: +49 6102 36 89-0	MEXICO Ph: +52-55-8850-0100	SWITZERLAND Ph: +41 43-45577 30	

Specifications are subject to change without notice.
Copyright (c) 2017 KEYENCE CORPORATION. All rights reserved.
160061E 1097-1 [691GB] Printed in Japan

